

**Polarization of Light:**

Polarization, in Physics, is defined as a phenomenon caused due to the wave nature of electromagnetic radiation. Sunlight travels through the vacuum to reach the Earth, which is an example of an electromagnetic wave. These waves are called electromagnetic waves because they form when an electric field that interacts with a magnetic field. In this article, you will learn about two types of waves, transverse waves, and longitudinal waves. You will also learn about polarization and plane polarized light.

*If such condition can be imposed on the vibration of a wave so that its vibration is confined along a particular direction or in a particular plane then it is called polarization.*

**Transverse Waves:**

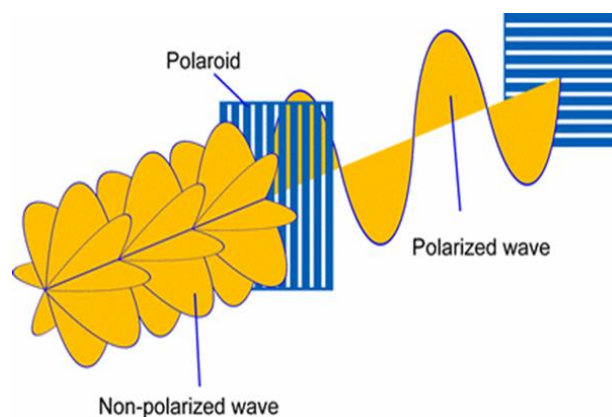
Transverse waves are waves, i.e., movement of the particles in the wave is perpendicular to the direction of motion of the wave.

Example 1: ripples in water, when you throw a stone.

Example 2: the motion of sound waves through the air.

Longitudinal waves are when the particles of the medium travel in the direction of motion of the waves.

Light is the interaction of electric and magnetic fields travelling through space. The electric and magnetic vibrations of a light wave occur perpendicularly to each other. The electric field moves in one direction and magnetic in another though always perpendicularly. So, we have one plane occupied by an electric field, the magnetic field perpendicular to it, and the direction of travel which is perpendicular to both. These electric and magnetic vibrations can occur in numerous planes. A light wave that is vibrating in more than one plane is known as unpolarized light. The light emitted by the sun, by a lamp or a tube light are all unpolarized light sources. As you can see in the image below, the direction of propagation is constant, but the planes on which the amplitude occurs is changing.

**Types of Polarization:**

Following are the three types of polarization depending on the transverse and longitudinal wave motion:

- Linear polarization
- Circular polarization
- Elliptical polarization

**Linear Polarization:** In linear polarization, the electric field of light is limited to a single plane along the direction of propagation.

**Circular Polarization:** There are two linear components in the electric field of light that are perpendicular to each other such that their amplitudes are equal, but the phase difference is  $\pi/2$ . The propagation of the occurring electric field will be in a circular motion.

**Elliptical Polarization:** The electric field of light follows an elliptical propagation. The amplitude and phase difference between the two linear components are not equal.

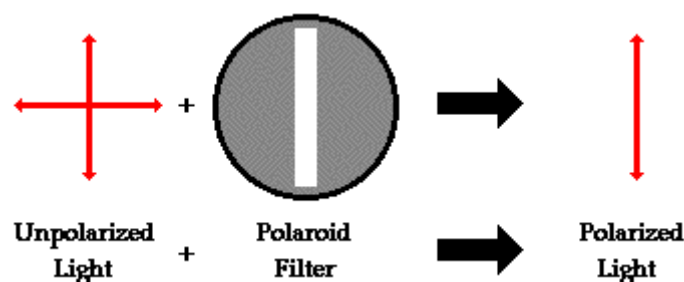
### Methods Used in the Polarization of Light:

There are a few methods used in the polarization of light:

- Polarization by Transmission
- Polarization by Reflection
- Polarization by Scattering
- Polarization by Refraction

#### **Polarization by Transmission:**

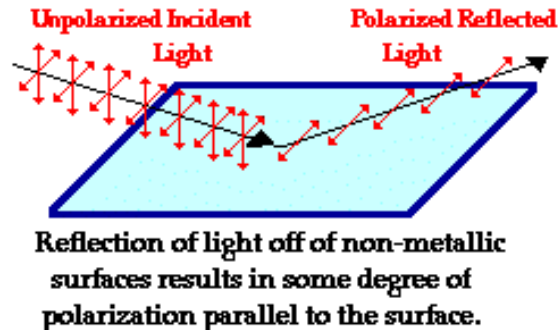
The most common method of polarization involves the use of a Polaroid filter. Polaroid filters are made of a special material that is capable of blocking one of the two planes of vibration of an electromagnetic wave. (Remember, the notion of two planes or directions of vibration is merely a simplification that helps us to visualize the wavelike nature of the electromagnetic wave.) In this sense, a Polaroid serves as a device that filters out one-half of the vibrations upon transmission of the light through the filter. When unpolarized light is transmitted through a Polaroid filter, it emerges with one-half the intensity and with vibrations in a single plane; it emerges as polarized light.



#### **Polarization by Reflection:**

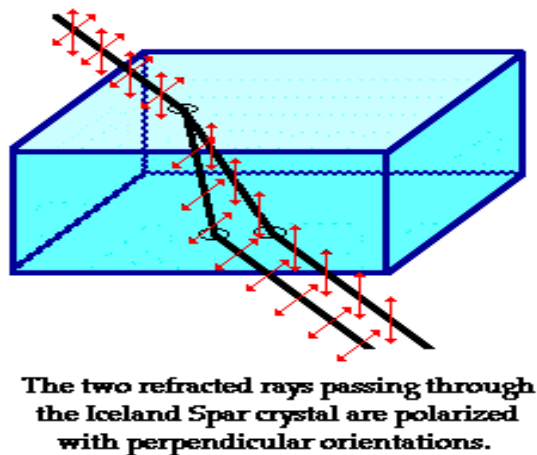
Unpolarized light can also undergo polarization by reflection off of nonmetallic surfaces. The extent to which polarization occurs is dependent upon the angle at which the light approaches the surface and upon the material that the surface is made of. Metallic surfaces reflect light with a variety of vibrational directions; such reflected light is unpolarized. However, nonmetallic surfaces such as asphalt roadways, snowfields and water reflect light such that there is a large concentration of vibrations in a plane parallel to the reflecting surface. A person viewing objects by means of light reflected off of nonmetallic surfaces will often perceive a glare if the extent of polarization is large. Fishermen are familiar with this glare

since it prevents them from seeing fish that lie below the water. Light reflected off a lake is partially polarized in a direction parallel to the water's surface. Fishermen know that the use of glare-reducing sunglasses with the proper polarization axis allows for the blocking of this partially polarized light. By blocking the plane-polarized light, the glare is reduced and the fisherman can more easily see fish located under the water.



### ***Polarization by Refraction:***

Polarization can also occur by the refraction of light. Refraction occurs when a beam of light passes from one material into another material. At the surface of the two materials, the path of the beam changes its direction. The refracted beam acquires some degree of polarization. Most often, the polarization occurs in a plane perpendicular to the surface. The polarization of refracted light is often demonstrated in a Physics class using a unique crystal that serves as a double-refracting crystal.



*The separation of a ray of light into two unequally refracted, plane-polarized rays of orthogonal polarizations, occurring in crystals in which the velocity of light rays is not the same in all directions, this phenomenon is called the double refraction of light.*

### ***Polarization by Scattering:***

Polarization also occurs when light is scattered while traveling through a medium. When light strikes the atoms of a material, it will often set the electrons of those atoms into vibration. The vibrating electrons then produce their own electromagnetic wave that is radiated outward in all directions. This newly generated wave strikes neighboring atoms, forcing their electrons into vibrations at the same

original frequency. These vibrating electrons produce another electromagnetic wave that is once more radiated outward in all directions. This absorption and reemission of light waves causes the light to be scattered about the medium. This scattered light is partially polarized.

### Polarization Applications:

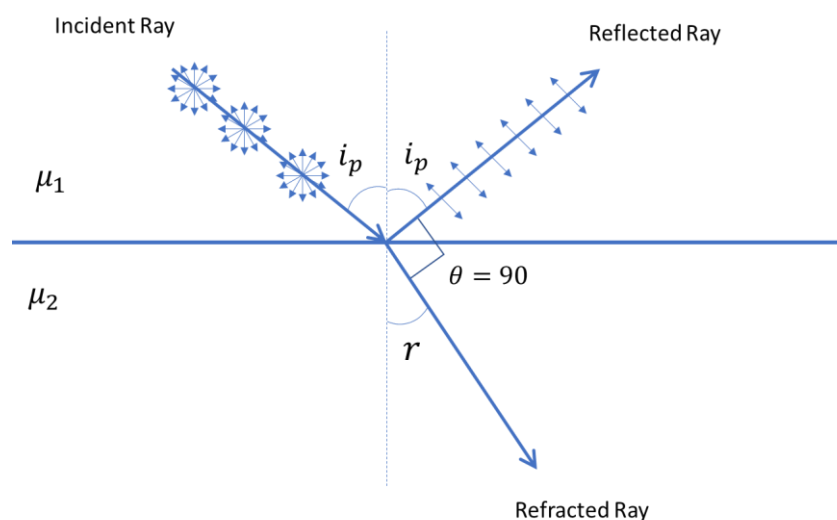
Following are the applications of polarization:

- Polarization is used in sunglasses to reduce the glare.
- Polaroid filters are used in plastic industries for performing stress analysis tests.
- Three-dimensional movies are produced and shown with the help of polarization.
- Polarization is used for differentiating between transverse and longitudinal waves.
- Infrared spectroscopy uses polarization.
- It is used in seismology to study earthquakes.
- In Chemistry, the chirality of organic compounds is tested using polarization techniques.

### Polarizing Angle or Brewster's Angle:

When unpolarized light is incident on the boundary between two transparent media, the reflected light is polarized with its electric vector perpendicular to the plane of incidence.

The polarization angle (also called Brewster's angle) is defined as the angle of incidence at which light is maximum polarized by reflection by the surface and refracted ray is slightly polarized.



### Brewster's Law:

*It states that the tangent of the angle of polarization is numerically equal to the refractive index of the medium.*

$$\tan i_p = \mu$$

According to the Snell's Law,

$$\mu = \frac{\sin i_p}{\sin r}$$

$$\Leftrightarrow \tan i_p = \frac{\sin i_p}{\sin r}$$

$$\Leftrightarrow \frac{\sin i_p}{\cos i_p} = \frac{\sin i_p}{\sin r}$$

$$\Leftrightarrow \cos i_p = \sin r = \cos \left( \frac{\pi}{2} - r \right)$$

$$\Leftrightarrow i_p = \frac{\pi}{2} - r$$

$$i_p + r = \frac{\pi}{2}$$

It seen from the diagram,

$$i_p + r + \theta = 180$$

$$\Leftrightarrow 90^\circ + \theta = 180^\circ$$

$$\theta = 90^\circ$$

So, at polarizing angle, the angle between the reflected ray and refracted ray from a surface is  $90^\circ$ .

**Problem-1:** What is the Brewster angle for air to glass transition? (Refractive index of glass = 1.5)

**Solution:** Given that,  $\mu = 1.5$

We have,  $\tan i_p = \mu = 1.5$

$$i_p = \tan^{-1}(1.5) = 56.31^\circ \quad (\text{Ans.})$$

**Problem-2:** If the refractive index of a polarizer is 1.9218. What will be the polarization angle and angle of refraction?

**Solution:** Given that,  $\mu = 1.9218$

We have,  $\tan i_p = \mu = 1.9218$

$$i_p = \tan^{-1}(1.9218) = 62.51^\circ \quad (\text{Ans.})$$

Again,  $i_p + r = 90^\circ$

$$r = 90^\circ - i_p = 90^\circ - 62.51^\circ = 27.49^\circ \quad (\text{Ans.})$$

**Some Definitions:**

- **Optic Axis:** If such a beam of ordinary unpolarized light is allowed to pass through a tourmaline crystal, only those vibrations which are parallel to a particular direction inside the crystal, called the optic axis of the crystal.
- **Plane of Vibration:** Ordinary or unpolarized light is many sided, the many sided being in respect of direction of vibration. The plane in which vibrations take place is known as the plane of vibration. The plane in which no vibrations occur is the plane of polarization which is at right angle to the plane of vibration.
- **Nicol Prism:** Nicol prism is an ingenious optical device for producing and analyzing plane polarized light and is based on the phenomenon of double refraction. It consists of two pieces of optically clear calcite or Iceland spar cemented together with Canada balsam in the shape of a prism.

**Malus' Law:**

we have seen that light, polarized by a tourmaline crystal, will pass through a second tourmaline crystal if the optic axis of this second crystal is parallel to that of the first crystal. The first crystal is referred to as the polarizer while the second crystal is known as the analyzer as it helps to detect polarized light since the eye alone cannot detect if light is polarized. When the optic axes of both the polarizer and the analyzer are parallel to each other, the intensity of the emergent polarized light is maximum.

Let this maximum intensity be  $I_0$ . As the analyzer is gradually rotated, the intensity of the transmitted light goes on decreasing until it becomes zero, when the optic axes of the two crystals are at right angles to each other. If the rotation of the analyzer is continued, then light gradually re-appears until the original intensity of the emergent light is restored when the analyzer has turned through  $180^\circ$  with respect to the polarizer i.e., the two optic axes again becomes parallel to one another. Consider the case when the analyzer has been rotated through an angle from its position of maximum intensity. The amplitude  $A$  of the plane polarized light incident on the analyzer can be resolved into two components perpendicular to each other. The component  $A \cos \theta$  is transmitted while the component  $A \sin \theta$  is either rejected or absorbed. Since the intensity is proportional to the square of the amplitude, the intensity  $I$  of the transmitted light at this position is given by –

$$I \propto A^2 \cos^2 \theta$$

$$\text{Also, } I_0 \propto A^2$$

$$\therefore \quad I = I_0 \cos^2 \theta$$

*This is known as Malus' cosine law according to which the intensity of the polarized light emerging from the analyzer is proportional to the square of the cosine of the angle between the polarizer and the analyzer.*

**Problem-3:** How will you orient the polarizer and the analyzer so that a beam of natural light is reduced to (i) 0.125 (ii) 0.25 (iii) 0.5 and (iv) 0.75 of its original intensity? [Answer:  $69^\circ.18'$ ,  $60^\circ$ ,  $45^\circ$ ,  $30^\circ$ ]

**Problem-4:** The critical angle for certain wavelength of light in the case of a piece of glass is  $40^\circ$ . Find the polarizing angle for glass. [Answer:  $57.3^\circ$ ]

**Problem-5:** Intensity of light through a polarizer and analyzer is maximum when their principal planes are parallel. Through what angle the analyzing Nicol must be rotated so that the intensity gets reduced to  $1/4$  of the maximum value. [Answer:  $60^\circ$  or  $120^\circ$ ]